

Jidu Yu

Postdoctoral Researcher

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RESEARCH INTERESTS

My research develops high-fidelity computational methods for geomaterials in extreme environments, where coupled thermal, hydraulic, mechanical and chemical (THMC) processes drive large deformation and failure.

- Particle and hybrid numerical methods: MPM, DEM, MPM-DEM, MPM-FVM
- Multiphysics (THM/C) modelling of multiphase granular media, e.g., frozen soils and hydrate-bearing sediments
- Climate-driven geohazards: permafrost thaw, rainfall-induced landslides, hydrate dissociation-induced instability
- Geomechanics in extreme environments: submarine, deep underground and extraterrestrial
- Machine learning-aided multiscale and multiphysics modelling
- 3D printing of extrusion-based materials: rheology and numerical modelling

ACADEMIC APPOINTMENTS

Feb. 2026 – present	Postdoctoral Researcher <i>Department of Civil and Environmental Engineering, University of California, Berkeley, USA</i> ADVISOR Prof. Kenichi Soga
Sep. 2025 – Jan. 2026	Postdoctoral Researcher <i>Department of Civil and Environmental Engineering, The Hong Kong University of Science and Technology</i> ADVISOR Prof. Jidong Zhao
Nov. 2024 – Apr. 2025	Visiting Ph.D. Researcher <i>Department of Civil and Environmental Engineering, University of California, Berkeley, USA</i> HOST Prof. Kenichi Soga
Jul. 2019 – Oct. 2019	Visiting Student <i>Department of Civil, Environmental and Geomatic Engineering, University College London, UK</i> HOST Prof. Yi Pik Helen Cheng

EDUCATION

Aug. 2021 – Aug. 2025	Ph.D. in Civil Engineering (Geotechnical) <i>The Hong Kong University of Science and Technology, Hong Kong</i> THESIS Multiscale Modeling of Coupled Thermo-Hydro-Mechanical Behavior in Granular Media SUPERVISOR Prof. Jidong Zhao
Sep. 2018 – Jul. 2021	M.Eng. in Hydraulic Structure Engineering <i>Hohai University, Nanjing, China</i> DISSERTATION Experimental and Theoretical Investigation of the Crushing Behavior of Granular Media SUPERVISOR Prof. Sihong Liu
Sep. 2014 – Jul. 2018	B.Eng. in Water Conservancy and Hydropower Engineering <i>Hohai University, Nanjing, China</i> DISSERTATION Experimental Study on the Compression Behavior of Crushable Granular Materials

RESEARCH GRANTS AND PROJECTS

2026 –	Research Grants Council of Hong Kong • General Research Fund (GRF) #16210526 A multiscale digital twin for sinkhole prediction: integrating MPM, CFD-DEM, and machine learning for urban geohazard assessment PI: Prof. Jidong Zhao Role: Co-I Funding: HK\$1,014,504
2025 –	National Natural Science Foundation of China A general-purpose multiscale, multiphysics computational platform for granular media PI: Prof. Jidong Zhao Role: Participant Funding: CN¥2.3 million
2025 –	Research Grants Council of Hong Kong • Theme-based Research Scheme #T22-607/24N A digital twin for enhancing coastal resilience against extreme storm surges in Hong Kong Project Coordinator: Prof. Jidong Zhao Role: Participant Funding: HK\$62.89 million
2023 – 2026	California Energy Commission • GFO-22-503 Performance-based monitoring and risk assessment tool for gas pipelines under natural forces PI: Prof. Kenichi Soga Role: Participant Funding: US\$4.41 million
2022 – 2024	Research Grants Council of Hong Kong • General Research Fund (GRF) #16206322 A combined physics-informed deep learning and multiscale modelling framework for simulation of thaw-induced landslide in permafrost PI: Dr. Shiwei Zhao (Co-PI: Prof. Jidong Zhao) Role: Participant Funding: HK\$802,220
2021 – 2023	Research Grants Council of Hong Kong • General Research Fund (GRF) #16211221 Multiscale modeling of thermo-hydro-mechanical (THM) behavior in granular sediments for gas hydrate recovery PI: Prof. Jidong Zhao Role: Participant Funding: HK\$911,317

HONORS AND AWARDS

2026	IJNAMG Top 10 Most-Cited Article & Top Viewed Article · Wiley	
2026	Research Travel Grant for Postdoctoral Fellow · HKUST	HK\$20,000
2024, 2025	RedBird Academic Excellence Award · HKUST	HK\$20,000 each
2024, 2025	Research Travel Grant for Postgraduate · HKUST	HK\$12,000 each
2024	Overseas Research Award · HKUST	HK\$60,000
2021 – 2025	Postgraduate Studentship · HKUST	HK\$225,120 per year
2022	Excellent Master's Dissertation in Hydraulic Engineering · China Association of Hydraulic Engineering Education	40 nationwide
2022	Excellent Master's Dissertation of Jiangsu Province · Jiangsu Provincial Degree Committee	150 province-wide
2022	Excellent Master's Dissertation · Hohai University	
2021	RedBird PhD Scholarship Program · HKUST	HK\$40,000
2021	Outstanding Postgraduate · Hohai University	
2021	Postgraduate "Science & Technology Star" · Hohai University	CN¥2,000 • top 20
2019, 2020	National Scholarship for Postgraduates · Ministry of Education of China	CN¥20,000 each
2018 – 2020	First Class Postgraduate Scholarship · Hohai University	CN¥12,000 each

2020	Postgraduate Research & Practice Innovation Program of Jiangsu Province · Jiangsu Province	CN¥20,000
2019	Outstanding Master Thesis Cultivation Program · Hohai University	CN¥20,000 • one per school
2019	First Class Prize, 9th “MathorCup” College Mathematical Modeling Challenge · Chinese Society of Optimization, Overall Planning and Economic Mathematics	
2018	Second Class Prize, 15th “Huawei Cup” China Postgraduate Mathematical Contest in Modeling · China Academic Degrees & Graduate Education Association	

PUBLICATIONS

Google Scholar: 337 citations · h-index 10 · i10-index 11 (Oct 2026)

First/corresponding-author papers in *JMPS* (2), *CMAME* (2), *IJNME* (1), *IJNAMG* (1), *Computers and Geotechnics* (1), *Construction and Building Materials* (1) and *JGGE* (1).

Bold: Jidu Yu * Corresponding author

Journal Articles

- J20 Chen X.Y., Zhao J.D. *, **Yu J.D.** (2026). The infiltration paradox: Pore-gas pressure controls on early slope failure during rapid hydraulic loading. *Engineering Geology*, 109124. doi:10.1016/j.enggeo.2026.109124
- J19 **Yu J.D.**, Zhao J.D. (2026). A virtual heat flux method for simple and accurate Neumann thermal boundary imposition in the material point method. *International Journal for Numerical Methods in Engineering*, 127(12), e70371. doi:10.1002/nme.70371
- J18 Liu Y.S., Shen C.M. *, Liu S.H., **Yu J.D.**, Liu Y.C., Li B.W. (2026). Refined constitutive modelling of widely graded granular soils incorporating fractional particle breakage. *Computers and Geotechnics*, 194, 107973. doi:10.1016/j.compgeo.2026.107973
- J17 Zhang C.X., Chen Y.N. *, **Yu J.D.**, Yang Z.X., Jardine R.J., Guo N. (2026). Stabilized semi-implicit material point method for hydro-mechanical coupled large deformation soil-structure interaction analyses. *International Journal for Numerical and Analytical Methods in Geomechanics*, 50(9), 3935–3954. doi:10.1002/nag.70314
- J16 **Yu J.D.** *, Zhao J.D. *, Liang W.J. (2026). Multiscale modeling of coupled thermo-hydro-mechanical-chemical behavior in hydrate-bearing sediment. *Journal of the Mechanics and Physics of Solids*, 210, 106512. doi:10.1016/j.jmps.2026.106512
- J15 **Yu J.D.**, Zhao J.D. *, Soga K., Zhao S.W., Liang W.J. (2026). A fully coupled THMC-MPM framework for modeling phase transition and large deformation in methane hydrate-bearing sediment. *Journal of the Mechanics and Physics of Solids*, 206, 106368. doi:10.1016/j.jmps.2025.106368
- J14 Liang W.J., Chandra B., **Yu J.D.**, Yin Z.Y. *, Zhao J.D. (2025). A total-Lagrangian material point method for fast and stable hydromechanical modeling of porous media. *International Journal for Numerical Methods in Engineering*, 126(19), e70135. doi:10.1002/nme.70135
- J13 **Yu J.D.**, Liang W.J., Zhao J.D. * (2025). Enhancing dynamic modeling of porous media with compressible fluid: A THM material point method with improved fractional step formulation. *Computer Methods in Applied Mechanics and Engineering*, 444, 118100. doi:10.1016/j.cma.2025.118100
- J12 **Yu J.D.**, Zhao J.D. *, Liang W.J., Zhao S.W. (2024). Multiscale modeling of coupled thermo-hydro-mechanical behavior in ice-bonded granular media subject to freeze-thaw cycles. *Computers and Geotechnics*, 171, 106349. doi:10.1016/j.compgeo.2024.106349
- J11 **Yu J.D.**, Zhao J.D. *, Zhao S.W., Liang W.J. (2024). Thermo-hydro-mechanical coupled material point method for modeling freezing and thawing of porous media. *International Journal for Numerical and Analytical Methods in Geomechanics*, 48(13), 3308–3349. doi:10.1002/nag.3794
- J10 **Yu J.D.**, Zhao J.D. *, Liang W.J., Zhao S.W. (2024). A semi-implicit material point method for coupled thermo-hydro-mechanical simulation of saturated porous media in large deformation. *Computer Methods in Applied Mechanics and Engineering*, 418, 116462. doi:10.1016/j.cma.2023.116462
- J9 Shen C.M., Liu S.H. *, Wang L.J., **Yu J.D.**, Wei H., Wu P. (2022). Packing, compressibility, and crushability of rockfill materials with polydisperse particle size distributions and implications for dam engineering. *Water Science and Engineering*, 15(4), 358–366. doi:10.1016/j.wse.2022.07.003

- J8 Shen C.M., Yu J.D.* , Liu S.H.* , Mao H.Y. (2021). A unified fractional breakage model for granular materials inspired by the crushing tests of dyed gypsum particles. *Construction and Building Materials*, 270, 121366. doi:10.1016/j.conbuildmat.2020.121366
- J7 Shen C.M., Liu S.H.* , Yu J.D., Wang L.J. (2021). A simple scale effect model for the volumetric behavior of rockfill materials. *International Journal of Geomechanics*, 21(3), 04020266. doi:10.1061/(ASCE)GM.1943-5622.0001939
- J6 Yu J.D., Shen C.M.* , Liu S.H., Cheng Y.P. (2020). Exploration of the survival probability and shape evolution of crushable particles during one-dimensional compression using dyed gypsum particles. *Journal of Geotechnical and Geoenvironmental Engineering*, 146(11), 04020121. doi:10.1061/(ASCE)GT.1943-5606.0002371
- J5 Wang T., Liu S.H.* , Feng Y., Yu J.D. (2018). Compaction characteristics and minimum void ratio prediction model for gap-graded soil-rock mixture. *Applied Sciences*, 8(12), 2584. doi:10.3390/app8122584

Journal Articles in Chinese

- J4 Yu J.D., Liu S.H.* , Shen C.M.* , Mao H.Y. (2022). Breakage and shape evolution of dyed gypsum particles under one-dimensional compression. *Journal of Hohai University (Natural Sciences)* (河海大学学报 (自然科学版)), 50(1), 110–116. (In Chinese) doi:10.3876/j.issn.1000-1980.2022.01.016
- J3 Wei H., Shen C.M., Liu S.H., Yu J.D. (2020). Experimental study on compression and crushing characteristics of coarse granular materials considering influence of gradations. *Journal of Hohai University (Natural Sciences)* (河海大学学报 (自然科学版)), 48(2), 182–188. (In Chinese) doi:10.3876/j.issn.1000-1980.2020.02.014
- J2 Yu J.D., Shen C.M.* , Liu S.H. (2020). Breakage and migration of dyed gypsum particles under one-dimensional compression. *Chinese Journal of Rock Mechanics and Engineering* (岩石力学与工程学报), 39(5), 1071–1079. (In Chinese) doi:10.13722/j.cnki.jrme.2019.0739
- J1 Yu J.D., Liu S.H.* , Wang T., Wei H. (2019). Experimental research on compaction characteristics of gap-graded coarse-grained soils. *Chinese Journal of Geotechnical Engineering* (岩土工程学报), 41(11), 2142–2148. (In Chinese) doi:10.11779/CJGE201911021

Manuscripts under Review

- R2 Liu X.Y., Gao G., Yu J.D., Meguid M.A., Zhang F.S. (2026). Micromechanical characterization of fracture evolution in confined jointed rocks using a 3D grain-based model. Submitted to *Rock Mechanics and Rock Engineering*.
- R1 Yu J.D., Chandra B.* , Wilkes C., Zhao J.D., Soga K. (2025). An elasto-viscoplastic thixotropic model for fresh concrete capturing flow-rest transition. Submitted to *Cement and Concrete Composites*. arXiv:2512.23364. arXiv:2512.23364

Conference Papers and Abstracts

- C7 Yu J.D., Zhao J.D.* (2026). High-fidelity multiscale framework for modelling permafrost thaw-related geohazards. *Proceedings of the 21st International Conference on Soil Mechanics and Geotechnical Engineering (IC-SMGE)*, 1743–1746, Vienna, Austria.
- C6 Yu J.D., Zhao J.D.* (2025). A multiphysics-multiscale MPM-DEM framework for modeling climate-driven geohazards in phase-change granular media. *IX International Conference on Particle-based Methods (Particles 2025)*, 20–22 October 2025, Barcelona, Spain.
- C5 Yu J.D., Zhao J.D.* (2025). Hybrid explicit-implicit THMC-MPM for modeling phase transition and large deformation in hydrate-bearing sediment. *The 2nd Workshop on Material Point Method & Extreme Deformation Problems (MPM 2025)*, 12–14 September 2025, Xi'an, China.
- C4 Yu J.D., Zhao J.D.* , Zhao S.W., Liang W.J. (2024). Thermo-hydro-mechanical coupled material point method for simulating the freeze-thaw behavior of porous media. *16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics (WCCM-PANACM 2024)*, July 2024, Vancouver, Canada.
- C3 Yu J.D., Zhao J.D.* , Zhao S.W., Liang W.J. (2024). Multiphysics simulation of freezing and thawing granular media using material point method. *GeoShanghai International Conference 2024*, May 2024, Shanghai, China. *IOP Conference Series: Earth and Environmental Science*, 1330, 012035. doi:10.1088/1755-1315/1330/1/012035
- C2 Yu J.D., Liang W.J., Zhao S.W., Zhao J.D.* (2023). A semi-implicit MPM for modeling the thermal effect in saturated porous media. *International Symposium on Innovations in Geotechnical Engineering towards Sustainability (IGES 2023)*, Hong Kong, China.

- C1 Yu J.D., Zhao J.D.* , Zhao S.W., Liang W.J. (2023). An explicit-implicit THM-coupled MPM for saturated porous media. *The 26th Annual Conference of HKSTAM 2023 in conjunction with the 18th Shanghai–Hong Kong Forum on Mechanics and Its Application*, Hong Kong, China, p. 30.

Thesis

- T1 Yu J.D. (2025). Multiscale modeling of coupled thermo-hydro-mechanical behavior in granular media. Ph.D. thesis, The Hong Kong University of Science and Technology, Hong Kong. doi:10.14711/thesis-hdl152441

INVITED TALKS

- Jan. 2026 **Multiphysics and multiscale modeling of geomaterials in large deformation: taking frozen soil and hydrate soil as examples**
College of Architecture and Civil Engineering, Beijing University of Technology, Beijing, China
- 2024 **Multiscale modeling the thermo-hydro-mechanical behavior of frozen soils**
College of Water Conservancy and Hydropower Engineering, Hohai University, Nanjing, China

TEACHING EXPERIENCE

- Teaching Assistant, The Hong Kong University of Science and Technology
- 2023 – 2025 **Undergraduate Final-Year Projects (2023–24, 2024–25)**
- Fall 2023 **CIVL 4750 Numerical Solutions to Geotechnical Problems**
- Spring 2023 **CIVL 3740 Geotechnical Analysis and Design**
- Spring 2022 **CIVL 4320 Structural Steel Design**

PROFESSIONAL SERVICE

- 2025 – **Journal reviewer**
Advances in Engineering Software; Ain Shams Engineering Journal; Chemical Engineering Science; Computational Particle Mechanics; Computer Methods in Applied Mechanics and Engineering; Computers and Geotechnics; Construction and Building Materials; Energy Technology; Engineering Geology; Finite Elements in Analysis and Design; Geoderma; International Journal for Numerical Methods in Fluids; International Journal of Geomechanics; International Journal of Heat and Mass Transfer; International Journal of Mining Science and Technology; Journal of Traffic and Transportation Engineering (English Edition); Measurement; Mechanics of Materials; Physics and Chemistry of the Earth, Parts A/B/C; Powder Technology; Simulation Modelling Practice and Theory; Tunnelling and Underground Space Technology